Khalil Lagha

M2 student in Electrical Energy Systems Design — power electronics & energy conversion · Université Grenoble Alpes

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SUMMARY

M2 student in Electrical Energy Systems Design (CSEE, UGA), **ranked 2nd of the M1 EEA cohort (15.43/20)**. Specialized in **power electronics**: **DC-DC converters (buck, boost)**, AC-DC rectification, **regulation-loop tuning (PI/PID)**, **efficiency optimization**, EMI compliance — validated in simulation (Matlab/Simulink) then on real hardware, across **five internships** from the lab (GIPSA-lab) to industrial sites. Target fields: **EV / powertrain**, **electrical grids**, **AI applied to power systems**. **Seeking: work-study contract (alternance) from September 2026 — 2–3-week company/university rotation.**

EXPERIENCE

Research Intern — GIPSA-lab, Grenoble (04/2026 - 07/2026, 3.5 months)

- Modeled a **10-cell PEMFC fuel-cell stack** in Matlab/Simulink and achieved **6%** dynamic error and **1%** static error against real measurements
- Tuned a **DC-DC boost converter** PI **regulation loop** to optimize efficiency; validated first in simulation, then on a real test bench

Intern — SLB (Schlumberger) (07/2025 - 07/2025, 1 month)

- Authored technical syntheses **in English** (directional drilling) in an international industrial environment

Power Electronics & Automation Intern — EURL Lagha (01/2024 - 01/2024, 15 days · 09/2024 - 09/2024, 1 month)

- Contributed to an **industrial DC-DC buck converter** — 50 V $\rightarrow$ adjustable 0–48 V, **25 A**
- Contributed to **4 transformer-rectifiers** feeding pipelines (electrolysis) — three-phase 400 V $\rightarrow$ 100 V transformer, rectification + capacitive filtering: **100 V DC / 80 A (**$\approx$ **8 kW)** output, ripple $<$ 5%, **EMI compliance**
- Co-designed the supervision HMI (**Siemens TIA Portal**, Ladder PLC)
- Analyzed a customer specification and sized the corresponding transformer; produced the electrical schematics in **EPLAN**

Grid Intern — Sonelgaz (03/2024 - 03/2024, 15 days, observation)

- Analyzed the full generation $\rightarrow$ distribution chain (**EHV 400/220 kV** $\rightarrow$ **HV 60 kV** $\rightarrow$ **MV 30/10 kV** $\rightarrow$ **LV 400/230 V**) and substation architecture; surveyed an MV/LV substation during a technical site visit

EDUCATION

- **M2 CSEE — Electrical Energy Systems Design**, Université Grenoble Alpes — from September 2026, apprenticeship
- **M1 EEA (EECS)**, Université Grenoble Alpes (2025–2026) — **ranked 2nd of the cohort, annual average 15.43/20**
- **Engineering cycle in Electrical Engineering**, École Nationale Polytechnique, Algiers (2021–2025) — 4 of 5 years completed; prep classes: among the top-ranked
- **Baccalauréat in Mathematics** (2021) — highest honors (16.93/20)

PROJECTS

- **Power grid analysis** (Matlab, solo) — developed **load flows from 3 to 118 buses** (PQ/PV/slack; IEEE 14/30/57/118 networks); benchmarked **3 numerical methods** — Gauss-Seidel, Newton-Raphson, fast-decoupled (accuracy vs computation time); studied **transient stability**
- **Field-oriented control (FOC) of an induction motor** (Matlab/Simulink) — implemented Park transforms, PID controllers and observers; tuned the **cascaded current- and speed-regulation loops**
- **Electrical machine sizing** (Python, team) — built a **Python (OOP, PyQt)** application: stator winding design, flux density via reluctance networks, **FEMM** finite-element validation, automatic parameter generation
- **Neural network from scratch (MLP)** (Matlab, open source) — hand-coded a multilayer perceptron and its **backpropagation** without any toolbox; published under the MIT license — github.com/KhalilEllahLAGHA/mlp-backpropagation-matlab
- Also: cascade control of a DC motor (three-phase thyristor bridge, Simulink); Siemens Step7 automation project (HMI + Ladder); Arduino line-follower robot (Poly Maze competition)

SKILLS

- **Power electronics**: DC-DC conversion (buck, boost), AC-DC (diode and thyristor rectification), DC-AC, AC-AC; converter sizing; PWM; filtering, ripple; EMI compliance; efficiency optimization
- **Control engineering**: regulation-loop tuning (PI/PID), LQR, observers, Kalman filter; field-oriented control (FOC); cascade control; system identification
- **Machines & grids**: electrical machines and transformers (analysis, design); power system analysis — load flow, Y/Z matrices, transient stability; MV/LV distribution
- **Industrial automation**: Siemens PLC (TIA Portal, Step7 — Ladder, SCL, FBD, SFC); HMI/SCADA; Grafcet
- **AI**: neural networks (backpropagation), genetic algorithms, fuzzy logic
- **Software**: Matlab/Simulink (Simscape Power Systems) · Python (OOP, PyQt) · C/C++ · LTspice · Proteus · EPLAN · AutoCAD Electrical · Siemens TIA Portal · Step7 (Ladder/SCL) · FEMM · SolidWorks · Arduino · Git/GitHub

SOFT SKILLS

Fast learner • Comfortable with tools/software • Problem solving • Rigorous • Organized — IEEE Student Branch ENP • VIC (ENP)

CERTIFICATIONS

- SLB — NEST & SIPP Level 2 (07/2025) — HSE, industrial-operations safety

LANGUAGES

French — C1 • **English — C2**